Classifying reported cardiovascular disease

BRFSS 2015: choices, experiments and results

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We selected histogram gradient boosting implemented in NumPy, combining Phase 18 diet representation, Phase 24 restricted imputation and 256 candidate features per node. The team reported **AIcrowd F1 0.455 and accuracy 0.883**. This final pipeline has 297 inputs. Its threshold-calibration F1 is 0.444071; an independent nested evaluation of this exact combination was not run.

Data and evaluation

The dataset contains 328,135 labelled rows, 321 predictors and 109,379 test rows. The target `_MICHD` records self-reported myocardial infarction or coronary heart disease. Positives make up 8.83% of training labels; 44.79% of predictor cells are blank. Always predicting negative gives 91.17% accuracy and F1 zero, motivating our focus on F1.

The initial stratified split (seed 20260918) gives 262,508 development and 65,627 local validation rows. Historical comparisons use three outer folds on development data (seed 20260919). Two inner folds within each outer training partition generate out-of-fold (OOF) predictions and select the F1 threshold; that threshold is applied to the excluded outer fold. Preprocessing, imputation, feature selection and sampling use only the training partition of each fit. Validation preserves all rows and prevalence. Later submission candidates instead calibrate thresholds on three OOF folds covering all labelled rows; those calibration scores are reported separately.

Methods and preprocessing

In `implementations.py`, we implemented the six required methods: linear regression with GD/SGD, least squares, ridge, logistic regression and L2 logistic regression. Least squares and ridge solve normal equations; linear losses use half-MSE and returned losses exclude regularisation. Historical linear baselines use median imputation, training-fitted scaling and age interactions. Logistic regression uses $\gamma = 0.2$, 4,000 updates and $\lambda = 0$; ridge uses $\lambda = 10^{-4}$ and inner-OOF logistic calibration.

Codebook cleaning. The CDC 2015 codebook guides types, units, response codes and exclusions. Removing identifiers, interview date, survey-design fields and redundant measurements leaves 295 source columns. Documented nonresponse codes become `NaN`; zero-quantity codes such as `ALCDAY5=888` and `PHYSHLTH=88` become zero. On retained development columns this recodes 1.10 million finite nonresponses and about 900,000 zero quantities. Histogram trees fit quantile bins and missing-value directions on training rows, then use logistic gradients and curvatures to update leaf predictions.

Final representation. Phase 18 removes six mixed-unit raw diet columns while retaining their existing CDC per-day equivalents, leaving 289 source columns. Phase 24 adds eight original missing-reason indicators and feature-only imputation: Naive Bayes fills documented unknown/refused responses in `INCOME2`, `PNEUVAC3` and `EMPLOY1`; ridge predicts missing height/weight and BMI is reconstructed when possible. Imputers and anchor bins are fitted within each training partition. Structural blanks remain missing. The combined representation has **297 columns**.

The selected model uses **200 trees, depth 5**, learning rate 0.05, at least 200 rows per leaf, 64 quantile bins, **256 candidates per node**, leaf L2 of 1, positive weight 1 and model seed 20260920. It retains the historical Phase 14 numerical implementation for exact reproduction.

Historical nested development evidence

Pipeline	F1	Precision	Recall	AP
Ridge	0.41951	0.33303	0.56665	0.39028
Logistic regression	0.42563	0.35728	0.52632	0.38775
Raw-input booster	0.44146	0.36663	0.55466	0.42783
Phase 14, cleaned 295 inputs	0.44216	0.36763	0.55457	0.42861
Phase 34, complete split search	0.44080	0.36200	0.56346	0.42878
MLP 64/32	0.42445	0.36072	0.51553	0.38778

Phase 14 exceeds logistic F1 by 0.01653. Cleaning exceeds raw-input F1 by 0.00070; its paired conditional 95% interval includes zero. Activity one-hot, exact discrete bins, restricted imputation, stronger regularisation, positive weights, two-seed averaging and selecting 100/60 columns did not improve the matched Phase 14 control. Random undersampling at 5:1/7:1 gives 0.43823/0.43982; random/hybrid 3:1 on corrected Phase 34 gives 0.43939/0.39353. These are historical comparisons, not a nested estimate for the final combined pipeline.

Full-training calibration and AIcrowd selection

For each candidate below, three stratified folds (seed 20260919) cover all 328,135 labelled rows. Each fold fits its own preprocessing and booster. Pooled OOF probabilities select the F1 threshold, breaking threshold ties by accuracy; a final fit then uses all labelled rows. **OOF F1 uses the same labels that selected the threshold and is a calibration score.** AIcrowd values are team-reported display scores at three decimals, recorded on 12–16 October 2026; they were not independently fetched from the platform.

Submission candidate	OOF F1	AIcrowd F1	AIcrowd accuracy
Phase 14 reference	0.442770	0.452	0.878
Phase 18 diet representation	0.444790	0.453	0.882
Phase 24 restricted imputation	0.443520	0.454	0.880
Phase 25 leaf L2=5	0.444429	0.453	0.883
Combined 18+24, 128 candidates, L2=1	0.443882	0.455	0.882
Combined 18+24, 128 candidates, L2=5	0.442655	0.453	0.883
Combined 18+24, 128 bins	0.443517	0.452	0.883
Combined 18+24, 256 candidates	**0.444071**	**0.455**	**0.883**
Separate changes to the selected 256-candidate configuration:			
300 trees	0.445145	0.453	0.883
297 candidates per node	0.442930	0.452	0.885
Minimum leaf size 400	0.443275	0.453	0.882
Maximum depth 4	0.442140	0.450	0.884
Native categorical splits on two activities	0.442326	0.452	0.885
80% stratified row sample per tree	0.443789	0.453	0.878
Leaf-wise growth, 32 leaves, depth 8	0.444142	0.452	0.881

The selected configuration ties the earlier combined model at the displayed F1 precision and has 0.001 greater displayed accuracy. Rounding prevents establishing an exact F1 improvement. All seven subsequent alternatives in the table have lower reported F1. The 300-tree model improves calibration F1 but loses 0.002 displayed AIcrowd F1, illustrating imperfect transfer from calibration to leaderboard feedback.

Three further count-feature candidates augment the 128-candidate combined base with positive-response and valid-response counts for functional limitations, other diagnoses, or both. They give calibration F1 0.443450, 0.443144 and 0.443352, below their base 0.443882. Their AIcrowd scores are unavailable; we do not infer them from local OOF results.

Interpretation and limitations

Split-search audit. Phase 34 adds an omitted candidate separating all observed values from missing responses. With Phase 14 features and parameters fixed, nested F1 changes from 0.442155 to 0.440803: 206 more true positives and 907 more false positives. Its paired conditional 95% interval [−0.00300, +0.00032] includes zero. The final submission keeps the historical search; results for it and the corrected algorithm remain distinct. These intervals condition on fixed fitted models and thresholds.

Disease timing. `HAREHAB1` asks about rehabilitation after a heart attack; all 620 recorded development responses belong to positives. Removing it from historical Phase 14 lowers F1 by 0.01112. It remains in the final model to classify reported disease. A model intended to predict before an event would require excluding it and auditing the timing of other responses. The score does not establish prospective clinical performance.

Repeated selection. Development folds were reused for many choices, and initial baselines consulted the local 20% partition. Later candidates use all labelled rows for calibration; AIcrowd feedback contributed to selection. Neither calibration F1 nor the selected leaderboard score is an untouched estimate of performance after this search. The historical nested results do not measure the exact combined submission.

Reproduction and final output

Install the dependencies, supply the official CSVs and run `python -m tools.prepare_data`. The selected implementation, preprocessing and threshold are fixed in `configs/final_model.json`. Train on all labelled rows and generate predictions with:

```
python run.py --data-dir data/raw/dataset --output submission.csv
```

The selected threshold is **0.22007772525099117**. The verified CSV has 109,379 rows: **13,895 positive and 95,484 negative**, header `Id,Prediction`, labels $-1/+1$ and IDs in official sample order. Its SHA-256 starts `d01c3949dea567dc`. Compact manifests, verification records and reported feedback are retained under `results/`; `docs/FINAL_REPORT.md` links the full evidence and calibration commands.